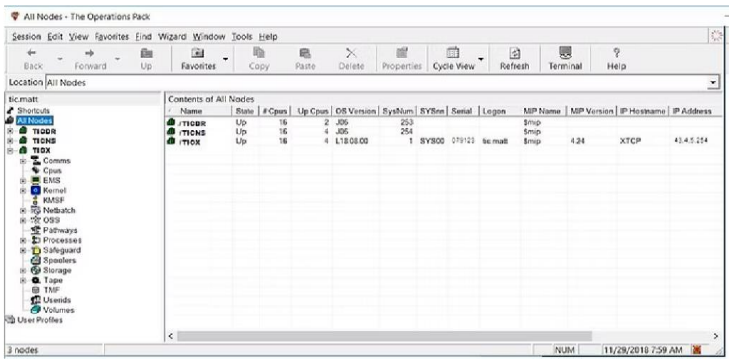
Using TOP with NonStop TCP/IP

TOP is a great GUI for just about everything you need to do on your NonStop system. One of the most helpful features is the ability to easily look at data communications information, especially as it pertains to CLIM IP, or “CIP”.

Let’s suppose that you want to see who is using TCP/IP on your NonStop system, and what programs are listening for incoming connections on a particular IP address, and which IP ports they’re using. Seems simple, right? But even if you do this on a frequent basis and can recall all of the SCF commands involved, it can be a tedious and time-consuming task:

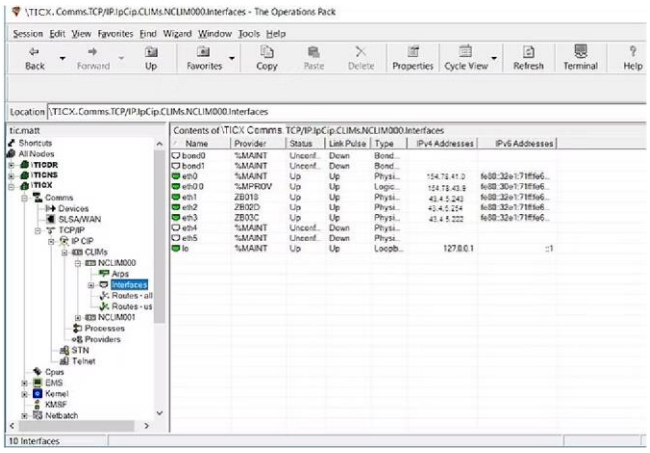
It is easy using TOP

Sign on to TOP and select your system. The system’s components appear in a familiar, WindowsExplorer-style interface.



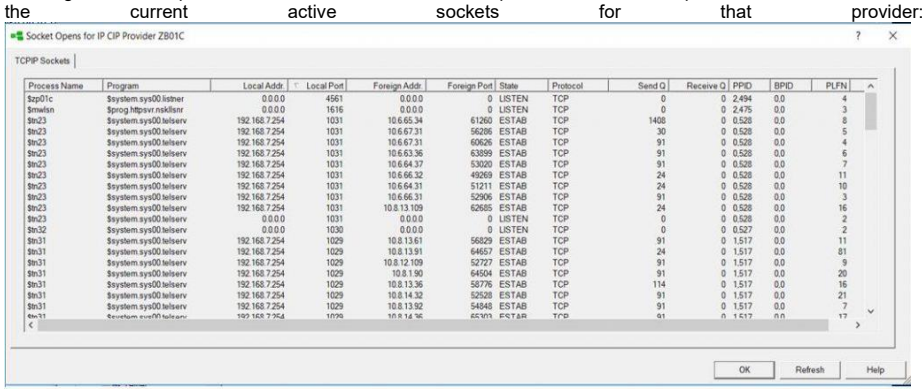
Navigate to “Comms”, “TCP/IP”, “IP CIP”

TOP now shows the available IP CLIMs. Select one and click on its “Interfaces”. You now see the CLIM’s interface information, including which providers are using them and which IP addresses are being served:

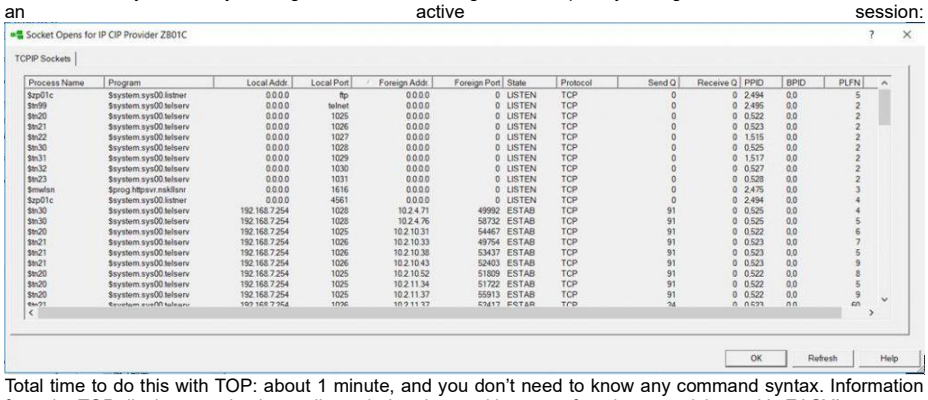


Now click on “Providers”. Top shows information on all the providers:

Next, right-click on a provider and select “Sockets” from the pull-down menu. TOP provides information for all of the current active sockets for that provider:



Sort it however you want by clicking on a column heading. For example, by Foreign IP Address, to see who has an active session:



Total time to do this with TOP: about 1 minute, and you don’t need to know any command syntax. Information from the TOP display can also be easily copied and pasted into your favorite spreadsheet. It’s EASY!

Doing it the long way using SCF

Now, the tedious way to do the same thing using SCF:

```
TACL> SCF
1-> assume process $ZZCIP
2-> status clim *

CIP Status CLIM

Name                    Config
State           Trace
NCLIM000                Yes
STARTED         OFF
NCLIM001                Yes
STARTED         OFF
SCLIM000                Yes
STARTED         OFF
SCLIM001                Yes
STARTED         OFF
SCLIM002                Yes
STARTED         OFF
```

Note that the command returns both IP CLIMs and Storage CLIMs. So, pick an IP CLIM and look at the detailed status. There's a LOT of information:

```
3-> status clim NCLIM000, detail
CIP Detailed Status CLIM \PAPEX.$ZZCIP.NCLIM000

Mode....................... IP
Multiprov................. ON
CLIM HW Config Status..... Configured
State..................... STARTED
Trace Status.............. OFF

Last Restart Time......... 11 Oct 2018, 13:58:22.000
CLIM Hostname............. NCLIM000
CIP SW Version............ T0853L03_01AUG2018_28AUG2018_DBF

Network SW Version........ T0691L03_01AUG2018_DBF_CLIM_G24
Storage SW Version........ T0830L03_01AUG2018_21AUG2018_DBF
Number of Socket Servers.. 1
Linux Version:
Linux 3.16.56-clim-4-amd64 #1 SMP Debian 3.16.56-clim-4
(2018-07-19)
Fabric Status:
CIPMON Mgmt Data
ZCM00 XY-1 XY-1
ZCM01 XY-1 XY-1
ZCM02 XY-1 XY-1
ZCM03 XY-1 XY-1

CLIM Maintenance Interface Status & IP Addresses:

lo             Sts:UP  Lkp:--    IPv4: 127.0.0.1
                                 IPv6: ::1
eth0           Sts:UP  Lkp:UP    IPv4: 154.78.41.0
                                 IPv6:
fe80::32e1:71ff:fe6a:6e90

Maintenance Provider Interface Status & IP Addresses:

lo             Sts:UP  Lkp:--    IPv4: 127.0.0.1
                                 IPv6: ::1
eth0:0         Sts:UP  Lkp:UP    IPv4: 154.78.43.9

Data Provider ZB018 Interface Status & IP Addresses:

Flg  Name      Status  LkP   Master / IP Family & Address
     lo        UP      --
                              v4: 127.0.0.2
                              IPv4: 127.0.0.1
                              IPv6: ::1
     eth1      UP      UP
                              IPv4: 43.4.5.243
                              IPv6:
fe80::32e1:71ff:fe6a:6e91

Data Provider ZB02D Interface Status & IP Addresses:

Flg  Name      Status  LkP   Master / IP Family & Address
lo        UP      --
                            IPv4: 127.0.0.2
                            IPv4: 127.0.0.1
                            IPv6: ::1
eth2      UP      UP
                            IPv4: 43.4.5.254
                            IPv6: fe80::32e1:71ff:fe6a:6e92

Data Provider ZB03C Interface Status & IP Addresses:

Flg  Name      Status  LkP   Master / IP Family & Address
lo        UP      --
                             IPv4: 127.0.0.2
                             IPv4: 127.0.0.1
                             IPv6: ::1
eth3      UP      UP
                             IPv4: 43.4.5.222
                             IPv6: fe80::32e1:71ff:fe6a:6e93

Data Provider ZTC0 Interface Status & IP Addresses:
Flg  Name      Status  LkP   Master / IP Family & Address
lo        UP      --
                             IPv4: 127.0.0.2
                             IPv4: 127.0.0.1
                             IPv6: ::1

Interface Failover Configuration and Status:
Flg  Name         Failover/(Assoc) Fovr  Current
Provider
NCLIM000.eth3                       --    NCLIM000.eth3
ZB03C

NCLIM000.eth1                       --    NCLIM000.eth1
ZB018

NCLIM000.eth2                       --    NCLIM000.eth2
ZB02D
```

Now you have the information needed to find out which IP addresses are associated with which interface on that CLIM. But who is using those addresses?

Next, find all of the CIPSAM processes – These are called “Providers” in CIP, and are the processes that your program accesses when it wants a connection. Each provider is associated with one or more CLIMs, and there may be more than one provider for a given CLIM:

```
4-> status provider *
CIP Status PROVIDER
Name        Status      TPStatus   Trace

ZB018       STARTED     Started    OFF
ZB01C       STARTED     Started    OFF
ZB02D       STARTED     Started    OFF
ZB02E       STARTED     Started    OFF
ZB03C       STARTED     Started    OFF
ZB03D       STARTED     Started    OFF
ZTC0        STARTED     Stopped    OFF
ZTC1        STARTED     Stopped    OFF
ZTCP0       STARTED     Started    OFF
ZTCP1       STARTED     Started    OFF
```

Well that wasn’t too hard. Next, find out which provider is associated with the IP address you’re interested in:

```
5-> status provider zb018,detail
CIP Detailed Status PROVIDER \TICX.$ZZCIP.ZB018
State..................... STARTED
TPStatus................. Started
Type..................... IPDATA
Family................... INET
Current Access........... ( 0, 1, 2, 3 )
Trace Status............. OFF
Interface Status & IP Addresses:
Name             Current          Status  LkP   Fovr  Index
lo                                UP      --    --
0x0000000001
                                  IPv4: 127.0.0.1
                                  IPv6: ::1

NCLIM000.lo                       UP      --    --
0x0000000001
                                  IPv4: 127.0.0.2
NCLIM000.eth1    NCLIM000.eth1    UP      UP    --
0x0000000007
                                  IPv4: 43.4.5.243
                                  IPv6:
fe80::32e1:71ff:fe6a:6e91
Interfaces Failover Configuration and Status:
F Home         Sts  LkP   Failover/(Assoc)Sts  LkP
Current       Fovr

NCLIM000.eth1  UP   UP                    --   --
NCLIM000.eth1 --
```

You might need to look at all the providers to find the one you're looking for. But great, now we know which CLIM, interface and IP address is driven by this provider. Now, enter another command to find all of the listeners and established sessions through the provider for that interface:

There may be a LOT of listeners and sessions, appearing in no particular order in the SCF output and, as you can see, it's not very easy to read. So next you copy the output from your screen (or log it to a file and copy that) into a spreadsheet, separate it into useful columns of information, and sort it to find what you're looking for.

That all will probably take you about 15 to 20 minutes if you know the commands by heart or maybe an hour or more if, like me, you have to read through the SCF manuals looking for them.

1 minute versus 15-20 minutes. Which way would you prefer?

You can use TOP to facilitate many other NonStop operations. Contact us to see all the ways it can help you.